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Joint optimization of logistics infrastructure investments and subsidies in a regional logistics network with CO_2 emission reduction targets

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ABSTRACT

This study proposes an optimization model that simultaneously incorporates the selection of logistics infrastructure investments and subsidies for green transport modes to achieve specific CO_2 emission targets in a regional logistics network. The proposed model is formulated as a bi-level formulation, in which the upper level determines the optimal selection of logistics infrastructure investments and subsidies for green transport modes such that the benefit–cost ratio of the entire logistics system is maximized. The lower level describes the selected service routes of logistics users. A genetic and Frank–Wolfe hybrid algorithm is introduced to solve the proposed model. The proposed model is applied to the regional logistics network of Changsha City, China. Findings show that using the joint scheme of the selection of logistics infrastructure investments and green subsidies is more effective than using them solely. Carbon emission reduction targets can significantly affect logistics infrastructure investments and subsidy levels.

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Introduction

Freight transportation is a major contributor to climate change, especially with global warming being attributed to various pollution emissions. Freight transportation is largely driven by the combustion of fossil fuels, mostly diesel fuel, which results in the emissions of greenhouse gases such as carbon dioxide (CO_2), nitrogen oxide, sulfur oxide, particulate matters, and air toxics (Qu et al., 2014). Freight transport is estimated to account for approximately 10% of energy-related carbon emissions (McKinnon et al., 2014). Therefore, implementing effective measures and policies to combat further environmental damage resulting from increased vehicular pollution emissions and developing sustainable, low-carbon regional transport and logistics systems are urgently needed.

Logistics network design and sustainable logistics management policies are two important aspects for developing a low-carbon regional logistics system. The former involves determining the optimal configuration of logistics infrastructure, whereas the latter mostly refers to subsidies for green transport modes or CO_2 emission taxes.

Determining optimal network configuration patterns and corresponding investment priorities for logistics infrastructure is an important objective in regional logistics network design (e.g., roads, railways, waterways, and logistics terminals)

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(Dablanc and Ross, 2012). Among all the configuration patterns of regional logistics network design, the configuration pattern based on a hub-and-spoke structure is the most widely applied (Wang, 2008). Alumur et al. (2012) investigated a hub location problem from the perspective of network design by jointly considering transportation cost and travel time and proposed a mixed integer programming formulation. Crainic et al. (2012) addressed a two-tiered freight distribution system in a large city to determine the optimal locations of logistics facilities.

The classic network design models based on a hub-and-spoke structure can efficiently describe the economies of scale by using transportation consolidation. However, such models cannot be used to characterize the interactive decision relationship among different shareholders. The freight network equilibrium models embedded with bi-level programming can capture the interactive decision relationships among different decision agents (logistics authorities, carriers, and shippers); these model involve seeking a transportation network supply configuration and a demand flow pattern that maximize a given objective function of a social type while satisfying the equilibrium constraint (Friesz et al., 1983; Harker and Friesz, 1986; Hurley and Petersen, 1994; Li et al., 2012b; Yang and Bell, 1998).

Harker and Friesz (1986) reviewed the major modeling techniques that had been applied in analyzing intercity freight network equilibrium and identified the shortcomings of spatial price equilibrium models. Guelat et al. (1990) investigated a multimode multiproduct network assignment model that had been used in Brazil to simulate flows, service levels, costs, and fuel consumption on the national multimodal freight network for multiple planning alternatives. Yamada et al. (2009) investigated the optimization problem of the multimodal freight transport network design and utilized a bi-level programming model for selecting a suitable set of actions from a set of alternatives (such as opening shipping lines, building ports, or adding new railways) with the objective of maximizing the benefit–cost ratio (BCR). Loureiro and Ralston (2001) developed a multi-commodity multimodal network optimization model designed to provide additional insights into the strategic stage of planning investments for intercity freight transportation networks; this model is solved with a heuristics algorithm combined with a column generation method. With the goal of minimizing total transportation costs, Meng and Wang (2011) developed an optimal programming model with equilibrium constraints for the intermodal hub-and-spoke network design problem to locate hubs and set highways, waterways, and railway corridors. Catalano and Migliore (2014) presented an optimal model for the design of logistics terminals based on the Stackelberg game, which aims to determine optimal location patterns and public shares in investments. This study revealed that transshipment and consolidation services should be concentrated at a few terminals to fully exploit economies of scale at the expense of the average travel time to reach logistics platforms. For a comprehensive review of multimodal freight network design, interested readers can refer to the references (Crainic, 2000; Crainic et al., 2009; SteadieSeifi et al., 2014).

The traditional logistics network design model mainly focuses on total costs or operator efficiency and only slightly considers external environmental costs. On the contrary, the green logistics network design problem focuses on improving logistics service efficiency, decreasing corresponding logistics costs, and reducing externalities while achieving a sustainable balance between economic, environmental, and social objectives (Dekker et al., 2012; Elhedhli and Merrick, 2012; McKinnon et al., 2012). Harris et al. (2011) assessed the effect of the traditional cost optimization approach to strategic modeling on overall logistics costs and CO_2 emissions by considering the supply chain structure and different freight vehicle utilization ratios. Their work indicated that the optimum design based on costs does not necessarily equate to an optimum solution for CO_2 emissions. Furthermore, a number of studies on logistics network design models integrate environment-related costs (e.g., GHGs) and the interactive decision relationship between logistics authorities and logistics users. The integrated logistics network optimization model is usually solved with bi-level programming; the upper level in this formulation searches for the optimal combinations of policy measures, whereas the lower level considers the behavior of network users when selecting logistics service routes (Nagurney and Toyasaki, 2003; Rudi et al., 2014; Zhang et al., 2015a,b).

Congestion pricing has been proposed as an efficient way to alleviate traffic congestion and to improve social welfare (Hammadou and Papaix, 2015; Wu et al., 2011). Yin et al. (2014) addressed the road toll pricing and capacity in a congested road network in a multi-criteria decision making framework by using a goal approach; this work provided an avenue for understanding the trade-offs among conflicting objectives and for designing a financially and environmentally sustainable transportation system. Dandotiya et al. (2011) addressed the joint optimization of freight rates and the terminal location problem to determine the optimal freight rate and the corresponding intermodal terminal location. King et al. (2014) examined the effect of road pricing on port freight activity in New York and New Jersey, USA. Their work showed that toll charges in the New York City metropolitan region may represent over 50% of the total costs for a short haul truck trip into or out of a maritime port depending on the location of the port facility and that road toll programs can impose non-trivial costs on truck trips to and from major regional freight centers.

Chen et al. (2014) investigated the coastal liner route design model for coastal intermodal networks subject to CO_2 emission reduction targets and state subsidy levels. The optimal model aims to determine ports of call, call sequences, ship types, and service frequencies and to minimize state subsidies for coastal shipping operators under a given carbon emission reduction target for the entire intermodal network. Bhadury and Eiselt (2012) developed a three-level location model to investigate the relationship among consumers, firms, and regional planners. The lowest level of the model describes the purchasing behavior of customers. At the second level, firms establish the optimal number of distribution centers to open with consideration of location costs, operation costs, and government subsidies. At the third and highest level, regional planners determine the optimal subsidies to maximize the total social welfare.

To the best of our knowledge, existing related studies integrating logistics infrastructure investments and subsidies for green transport modes are still scarce. The present work aims to fill this gap by focusing on the joint optimization of logis-

tics infrastructure network investments and subsidies for green transport modes under specific CO_2 emission reduction targets.

First, a bi-level programming model is proposed to address the simultaneous optimization problem of the selection of logistics infrastructure investments and subsidies for green transport modes subject to specific CO_2 emission reduction targets. The upper level of the model is used to maximize the ratio of the reduced total generalized logistics cost and the investment cost incurred for implementing the actions, whereas the lower level is formulated as a user equilibrium problem for logistics service route choices over a multi-mode regional logistics network. Second, a genetic algorithm (GA)-based solution is adapted to solve the proposed bi-level model. The effects of the investment and subsidy combination plan are also compared.

The rest of this paper is organized as follows. The basic components of the model, including the regional logistics network representation and model assumptions, are described in Section 'Basic considerations'. The model formulation is presented in Section 'Model formulation', and the solution algorithm is discussed in Section 'Solution algorithm'. A numerical example for illustrating the model applications is presented in Section 'Case study'. Finally, the drawn conclusions are elaborated in Section 'Conclusion and future studies'.

Basic considerations

Network representation

To model regional logistics services, we present in Fig. 1a–d the logistics demand network, logistics service network, service route, and physical logistics network, respectively. Fig. 1a shows the different types of logistics demands (such as industrial, commercial, and agricultural logistics) between a given logistics origin–destination (OD) pair. These demands are served by the logistics service network, as shown in Fig. 1b and c.

As shown in Fig. 1d, the logistics physical network is composed of a set of logistics nodes (logistics parks, distribution centers, and freight terminals) and a set of logistics links or arcs that represent physical links, such as road segments, rail tracks, river segments, and sea lines. We denote a logistics physical network as $G_p = (N_p, A_p)$, where N_p is the set of logistics nodes (or transfer nodes) and A_p is the set of logistics links. All transfers take place at logistics transfer nodes. In this work, two types of logistics transfer nodes are considered: one is a logistics park with economies of scale effects, and the other is a general transfer node with a small capacity, such as a distribution center.

To facilitate the construction of the model, we introduce a virtual arc that represents a logistics transfer activity (i.e., change of transportation mode). For example, Fig. 2 represents a logistics service from node A to node B that is delivered via transfer from a railway to a highway at transfer node H. For ease of presentation, we denote $N_s^h \subset N_s$ as the set of logistics transfer nodes and $A_s^h \subset A_s$ as the set of virtual logistics transfer arcs.

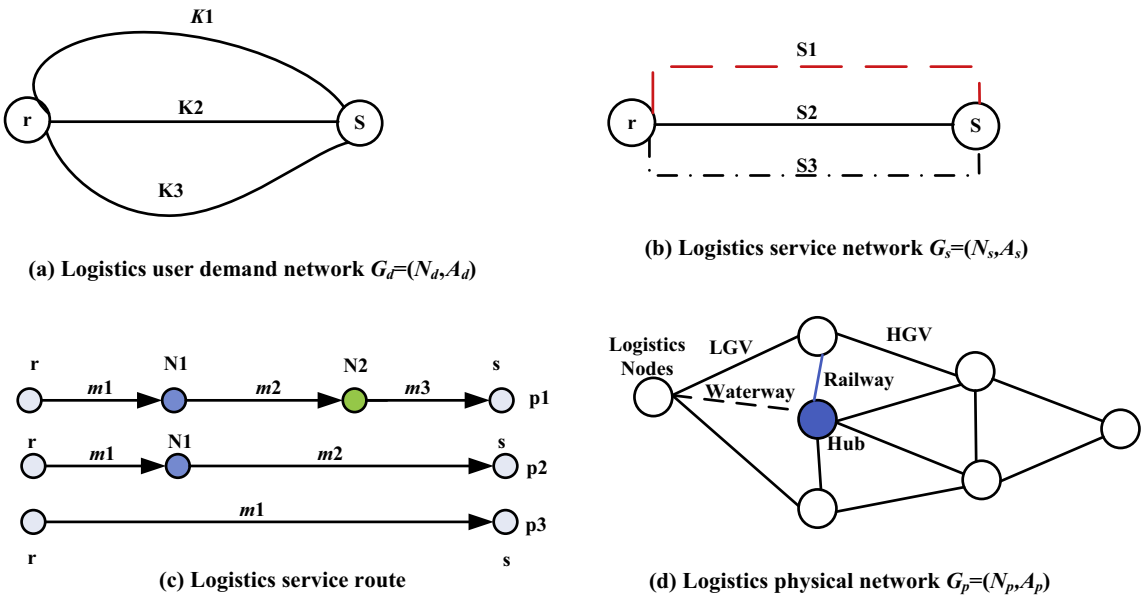


Fig. 1. Network representation of a regional logistics system.

Assumptions

To facilitate the presentation of essential ideas without the loss of generality, the following basic assumptions are made.

A1 The planning period is assumed to be one week. Thus, the proposed model is mainly used for strategic planning and policy evaluation.
A2 In the regional logistics system, the logistics infrastructure investment and subsidy for railways or waterways are determined by the logistics authority.
A3 The industry logistics demand points are distributed in the outer ring of the city, whereas the commercial logistics demand points are mainly distributed in the inner city.
A4 Logistics users choose their logistics service routes on the basis of their own perceptions of service disutility. The disutility of a logistics service is measured by transport time, transport cost, and subsidy for green transport mode.
A5 Logistics users can use a single (pure) mode or a combination of several modes (combined mode). The single or pure mode includes heavy goods vehicles (HGVs), light goods vehicles (LGVs), railways, and waterways.

Notations

Sets

M	$M = \{1, 2, 3, 4\}$ set of all transport modes in logistics networks, "1", "2", "3",and "4", respectively, represent the HGVs, LGVs, railways, and waterways
W	set of all origin–destination (O–D) pairs in the logistics network
$\mathbf{R}_w$	set of all service routes between OD pair $w \in W$
N	set of logistics nodes
K	$K = \{1, 2\}$ set of investment of logistics infrastructure option, "1" and "2", respectively, represent the expanding the capacity of existing logistics infrastructures and adding new ones respectively
A	set of links
$\mathbf{A}^0$	set of existing links without modification
$\mathbf{A}^1$	set of links after the associated improvement action is implemented

General variables

q_w	the demand between OD pair $w \in W$ (tons/week)
t_a^m	transport time on arc $a \in A_s^M$ by transport mode m (h)
μ_a^m	disutility on arc $a \in A_s^M$ by transport mode m ($)
δ_{ar}^m	indicator variable, equals 1 if link a is on route r by mode m, and 0 otherwise
h_{wr}^m	freight flow served by combined transport mode m on route $r \in R_w$ between OD pair $w \in W$ (tons/week)
v_a^m	freight flow on logistics service arc $a \in A$ by transport mode m (tons/week)

Decision variables (logistics authority)

x_a^k	0–1 variable that is equal to 1 if the logistics infrastructure investment option k over link a is selected and to 0 otherwise
$\mathbf{X}$	vector of 0–1 variable $x_a^k, \mathbf{X} = (x_a^k, \forall a \in A, k \in K)$
y_a^m	size of the subsidy for green transport mode m over link a
$\mathbf{Y}$	vector of variable $y_a^m, \mathbf{Y} = (y_a^m, \forall a \in A, m \in M)$

Constants

q_w^m	potential demand between OD pair $w \in W$ by transport mode m (tons/week)
t_a^{m0}	free flow transport service time on arc $a \in A$ by transport mode m (h)
t_{md}	average shift interval for transport mode m (h)
l_a^m	length of arc $a \in A$ by transport mode m (km)
c_a^m	fare of unit turnover on arc $a \in A$ served by mode m ($/ton km)
Cap_a^m	service capacity on arc $a \in A$ by mode m (tons/km)
I_a^k	the fixed cost of candidate logistics infrastructures project k over link a ($/ton)
τ_{vot}	value of time (VOT) ($/ton h)
e_a^m	average CO_2 emission per unit turnover on arc $a \in A$ by transport mode $m \in M$ (kg/ton km)
R_E	the expected minimum CO_2 emission reduction target (%)
B	the available investment budget ($)
S_a^m	the maximum value of the subsidy for transport mode m on arc $a \in A$

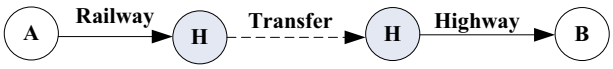


Fig. 2. Representation of a virtual transfer arc.

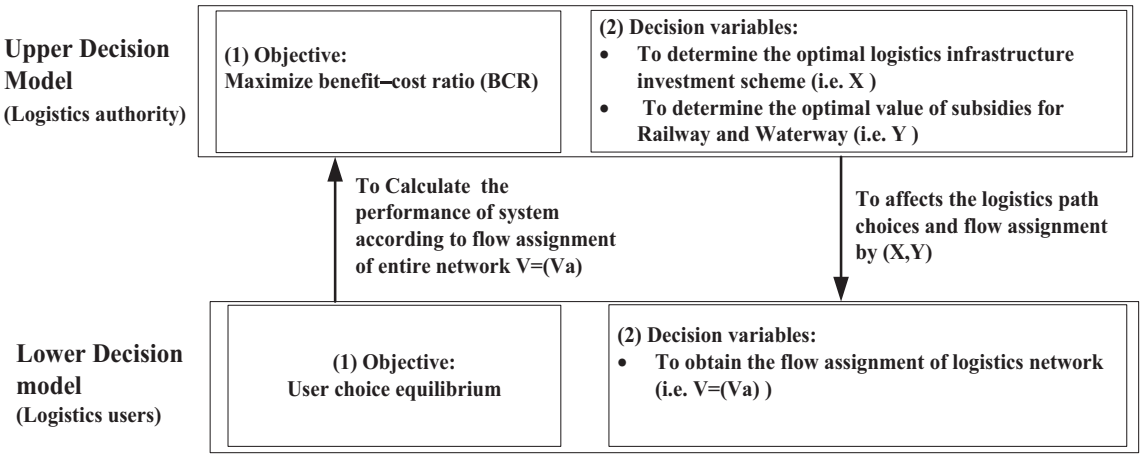


Fig. 3. Decision framework of green logistics system.

Model formulation

A logistics system comprises two interrelated players, namely, a logistics authority and users (i.e. carriers). The goal of a logistics authority is to choose the optimal investment combination with subsidies such that the BCR of the entire logistics system is maximized subject to CO_2 emission reduction targets. The investment options to be considered by the model include the addition of new physical links to the network, the improvement of existing links, and the location of logistics terminals at specified nodes of the network. The decision of the logistics authority affects the service choices of logistics users. Logistics users choose their routes and modes on the basis of their own perception of logistics service disutility. Such perception further affects the decision of the logistics authority. The interaction between the logistics authority and the users can be formulated as a bi-level model, as shown in Fig. 3. We formulate the decisional behavior of the logistics authority and users below.

The joint optimization of investments and subsidies in regional green network design is a bi-level programming model that consists of two sub-models, as shown in Fig. 3. The upper-level decision model is an integer programming model. This model determines the optimal combination of selected green infrastructure investments and subsidies for selected green logistics service routes to maximize the total BCR of the logistics system while meeting carbon emission reduction targets.

The lower-level decision model is the UE assignment model. This model fulfills three functions:

(a) Assigning the logistics OD to the logistics network $G = (V,A)$.
(b) Estimating the carbon emission of the entire logistics system.
(c) Returning the results to the upper-level decision model to calculate the BCR for government actions related to investments and subsidies.

The upper-level and lower-level models are discussed in detail below.

Specification of the upper-level decision model

The upper-level model is described as follows:

$$\max Z_U = \frac{\sum_{a\in A^0}\sum_{m\in M}\mu_a^m(v_a^m)v_a^m - \sum_{a\in A^1}\sum_{m\in M}\mu_a^m(v_a^m,x_a^m,y_a^m)v_a^m}{\sum_{a\in A^0}\sum_{K\in K}x_a^k l_a^k + \sum_{a\in A^1}\sum_{m\in M}y_a^m v_a^m l_a^m} \tag{1}$$

subject to

$$\frac{E_s^0(\mathbf{V}^0) - E_s^1(\mathbf{V}^1)}{E_s^0(\mathbf{V}^0)} \geqslant R_E \tag{2}$$

$$\sum_{a\in A^0}\sum_{m\in M} x_a^m l_a^m + \sum_{a\in A^1}\sum_{m\in M} y_a^m v_a^m l_a^m \leqslant B \tag{3}$$

$$0 \leqslant y_a^m \leqslant s_a^m \quad \forall a \in A,\ m \in M \tag{4}$$

$$x_a^k = \{0,1\} \quad \forall a \in A,\ k \in K \tag{5}$$

where

$$E_s^0(V^0) = \frac{\sum_{a\in A^0}\sum_{m\in M} e_a^m l_a^m v_a^m}{\sum_{a\in A^0}\sum_{m\in M} v_a^m l_a^m} \tag{6}$$

$$E_s^1(V^1) = \frac{\sum_{a\in A^1}\sum_{m\in M} e_a^m l_a^m v_a^m(x_a^m, y_a^m)}{\sum_{a\in A^1}\sum_{m\in M} v_a^m(x_a^m, y_a^m) l_a^m} \tag{7}$$

The objective function (Eq. (1)) maximizes the BCR of the logistics system, which is the ratio of the reduced generalized total logistics cost to the investment and subsidy cost required for implementing the set of green logistics initiatives. In Eq. (1), the numerator represents the reduced generalized total freight cost, which is computed as the difference between the total generalized costs incurred with and without the implementation of green initiatives, whereas the denominator represents the combined total of the logistics infrastructure investments and subsidies for green transport modes. Constraint (2) ensures that the expected minimum CO_2 emission reduction target R_E is realized. Constraint (3) implies that the total expenditure of implementing infrastructure projects cannot exceed the budget limits. Constraint (4) states the limited range of subsidies for green transport mode. Constraint (5) represents the decision to select a logistics infrastructure investment option k over link a. Eqs. (6) and (7) represent the CO_2 emissions of a unit turnover of the logistics system with and without the implementation of investment actions, respectively.

Specification of the lower-level decision model

Considering the difference in the attributes of various transportation modes, the logistics service time on link a for different modes should be estimated by using diverse functions, as shown in Eq. (8). For HGVs or LGVs, a US Bureau of Public Roads-type function can be used to estimate service time (Lam et al., 1999; Zhang et al., 2015a,b). For railways or waterways, the average transport service time and departure interval time should be considered, i.e.,

$$t_a^m(v_a^m) = \begin{cases} t_a^{m0}\left(1 + 0.15\left(\frac{v_a^m}{Cap_a^m}\right)^4\right), & m = 1,2 \\ t_a^{m0} + t_{md}\left\lceil \frac{\max(v_a^m - Cap_a^m, 0)}{Cap_a^m} \right\rceil, & m = 3,4 \end{cases} \quad \forall a \in A_s \tag{8}$$

where t_a^{m0} is the free-flow transport service time, and t_{md} is the average shift interval for transport mode m.

According to A4, the disutility of logistics users consists of the logistics service time, transportation cost, and subsidy, which can be expressed as

$$u_a^m(v_a^m, x_a^k, y_a^m) = c_a^m(v_a^m, x_a^k, y_a^m)(1 - y_a^m) + \tau_{vot} T_a^m(v_a^m, x_a^k, y_a^m), \quad \forall a \in A,\ m \in M,\ k \in K \tag{9}$$

For certain network capacity enhancements and subsidy levels for green transport mode, the equilibrium link flow pattern can be obtained by solving the following mathematical programming problem (Bhadury and Eiselt, 2012).

$$\min\ Z_L(v_a^m, x_a^k, y_a^m) = \sum_{a\in A}\sum_{m\in M}\int_0^{v_a^m} \mu_a^m(\omega, x_a^k, y_a^m) d\omega \tag{10}$$

$$\sum_{r\in R_w} h_{wr}^m = q_w^m \quad \forall w \in W,\ m \in M \tag{11}$$

$$\sum_{w\in W}\sum_{r\in R} h_{wr}^m \delta_{ar}^m = v_a^m \quad \forall a \in A \tag{12}$$

$$v_a^m \geqslant 0, \quad \forall a \in A,\ m \in M \tag{13}$$

Eq. (11) is the OD demand conservation constraint. Eq. (12) is the interrelation between link flow and route flow. Eq. (13) is the nonnegative constraint on route flow. Using the UE model in Eqs. (10)–(13), we can obtain the equilibrium link flow pattern $\mathbf{V} = (v_a^m, \forall a \in A, m \in M)$.

Solution algorithm

In this section, we solve the bi-level decision model by developing a solution algorithm based on a GA and the Frank–Wolfe hybrid algorithm (Geoffrion, 1969; Li et al., 2012a; Nguyen et al., 1998). The process of the above hybrid algorithm is described as follows.

***Step 1*. Initialization.**

(1) Set the GA parameters: population size (*pop*), crossover probability (P_c), and mutation probability (P_m).
(2) Set a stopping criterion and a predefined number of generations (Max_gen).
(3) Set $n = 0$, and create the initial population $pop_n = \{G_k^n | k = 1, 2, 3, \ldots, p\}$, in which p is the size of the population, individual G_k^n is the k-th decision scheme in the n-th generation; that is, $G_k^n = \{X_k^n, Y_k^n\}$ is the decision vector, with X_k^n representing the selected investments and Y_k^n representing the subsidies for railways and waterways.

***Step 2*. Inner loop operation** (solve the lower-level UE mode–flow assignment)**.**
For a given logistics authority decision $G_k^n = \{X_k^n, Y_k^n\}$, obtain the flow v_a^m over link a by transport mode m by using the Frank–Wolfe algorithm.
***Step 3*. Calculate the fitness value of G_k^n.**
Compute the BCR value $(Z_U)_k^n$ on the basis of Eqs. (1), (6), and (7).
Let $fit_k^n = (Z_U)_k^n$. For each G_k^n of pop_n, calculate the fitness value fit_k^n on the basis of $(Z_U)_k^n$.
***Step 4*. Sorting.**
Sort the individual G_k^n of the population pop_n in descending order according to the value fitness fit_k^n.
***Step 5*. Selection process.**
The selection process involves spinning the roulette wheel *pop_size* times each time we select a single chromosome for a new population in the following way.

(1) Calculate the evaluation function.

We can adopt the so-called rank-based evaluation function as follows:
$Eval(V_i) = a(1-a)^{i-1}, \forall i = 1, 2, 3, \ldots, pop_size$. Let $a = 0.05$.
We mention that $i = 1$ denotes the best individual, whereas $i = pop_size$ denotes the worst individual, as well as $\sum_{i=1}^{pop_size} Eval(V_i) = 1$.

(2) Calculate the cumulative probability $Cum(i)$ for each chromosome V_i, that is, $Cum(0) = 0, Cum(i) = \sum_{j=1}^{i} Eval(V_j), \forall i = 1, 2, 3, \ldots, pop_size$.
(3) Generate a random real number r in $(0, Cum[pop_size]]$.
(4) Select the i-th chromosome $V_i (1 \leqslant i \leqslant pop_size)$ such that $Cum(i-1) < r \leqslant Cum(i)$.
(5) Repeat (2)–(4) *pop_size* times to obtain the *pop_size* copies of chromosomes.

In this process, the best chromosomes are duplicated several times, the average chromosomes stay the same, and the worst ones die off.
***Step 6*. Implement crossover and mutation.**
Implement crossover and mutation operations to obtain a new population pop_n and set $n = n + 1$. The detailed description of the crossover and mutation operations is presented in Section 'Crossover and mutation method'.
***Step 7*. Conduct a termination check for the outer loop operation.**
If ($n <$ max_gen), then stop the calculation; otherwise, proceed to step 2.
***Step 8*. Output the final optimal solution.**
The flow chart of the above algorithm is shown as Fig. 4.

Key element design of GA

The performance of a hybrid GA algorithm is largely influenced by the design of genetic operators and the value of the parameters. In this section, we explain the three key elements of the GA used in this work: coding, crossover and mutation methods.

Coding method
The chromosome of the individual in the GA is composed of two parts, as shown in Fig. 5.
Part 1 represents the selection of the candidate logistics infrastructure. That is, a value of 1 equates to the condition in which the candidate logistics infrastructure is indeed selected; otherwise, the value is 0. As shown in Part 2, the values of the subsidies for railways and waterways are y_1 and y_2, respectively.

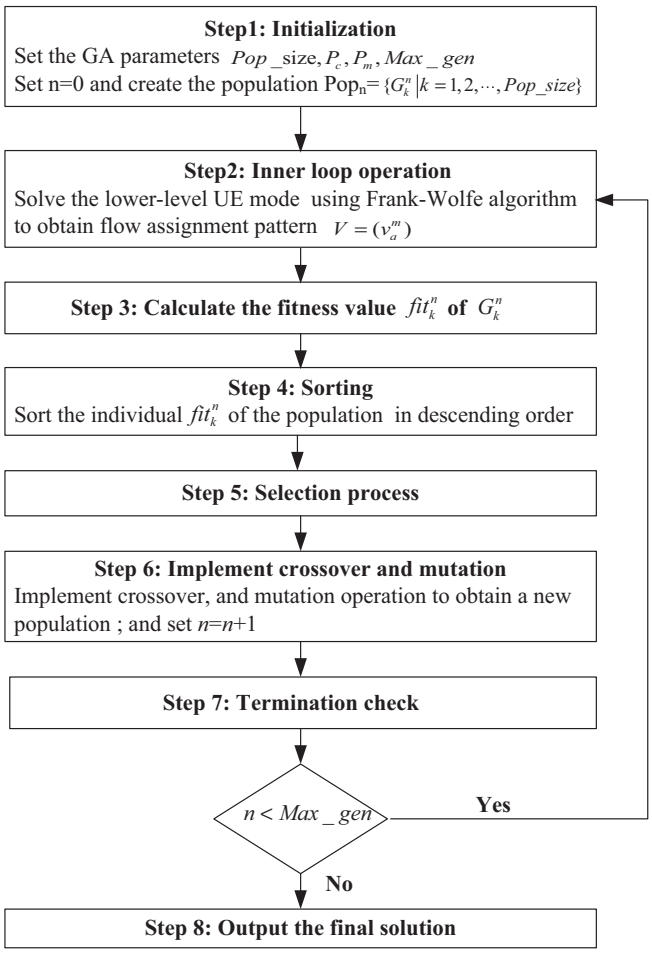


Fig. 4. The flow charts of algorithm.

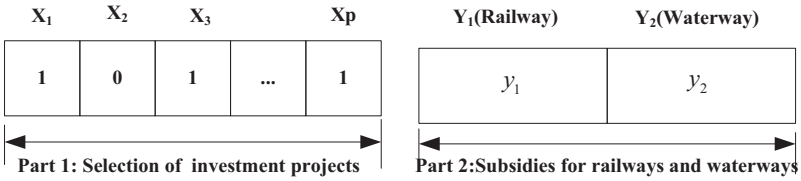


Fig. 5. Chromosome of the GA.

Crossover and mutation method

(1) Crossover operator

First, we generate a random real number r in [0,1]. Second, we select the given chromosome for a crossover operation if $r \leqslant P_c$. We repeat this operation *pop_size* times and produce $P_c \cdot pop_size$ parents on average.

In this study, we implement a crossover operation for the two parts of the chromosome. The two-point crossover method is adopted for Part 1. The crossover operation in Part 2 is a convex combination of two corresponding parents. For each pair of parents Y_1 and $\boldsymbol{Y}_2$, we can obtain two children C_1 and C_2 with a crossover operator as follows:

$$C_1^{(1)} = \lambda_1 y_1^{(1)} + \lambda_2 y_1^{(2)} \text{ and } C_1^{(2)} = \lambda_2 y_2^{(1)} + \lambda_1 y_2^{(2)}, \text{ where } \lambda_1, \lambda_2 > 0; \lambda_1 + \lambda_2 = 1.$$

$$C_2^{(1)} = \lambda_1 y_1^{(1)} + \lambda_2 y_1^{(2)} \text{ and } C_2^{(2)} = \lambda_2 y_2^{(1)} + \lambda_1 y_2^{(2)}, \text{ where } \lambda_1, \lambda_2 > 0; \lambda_1 + \lambda_2 = 1.$$

We provide a simple example that consists of six candidate logistics parks, as shown in Fig. 6.

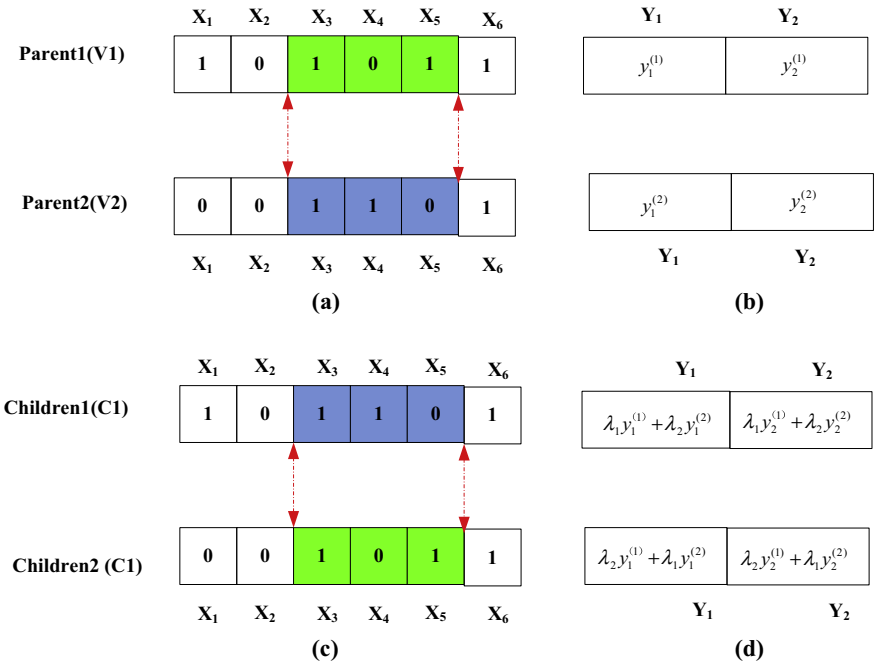


Fig. 6. Representation of the crossover operation.

(2) Mutation operator

We perform the mutation on the set of offspring produced by the crossover operations. Generating a random real number r in [0, 1], we select the given chromosome for mutation if $r < P_m$. Let a parent $\mathbf{V}$ for mutation, denoted as a vector $\mathbf{V} = (\mathbf{X}, \mathbf{Y})$, be selected.

For the first part (i.e., vector X), the two-point mutation method is adopted. In this method, the genome bits of the middle segments (between position 1 and position 2) are inverted (i.e., if the genome bit is 1, it is changed to 0, and vice versa), as shown in Fig. 7a.

For the second part (i.e., vector Y), its mutation operator is described, as shown in Fig. 7b. Given a large positive number M (e.g., M = 10,000) and a random direction d (e.g., in two-dimensional space R^2, a feasible direction d = (0.112, −0.807)).

If the vector $\mathbf{Y} + M \cdot \mathbf{d}$ is feasible, we use $\mathbf{Y} + M \cdot \mathbf{d}$ to replace $\mathbf{Y}$; otherwise, we set M as a random number between 0 and M until the vector becomes feasible. If the use of the above-mentioned process does not lead to a feasible solution after a predetermined number of iterations *Max_iter*, then we consider $M = 0$. Thus, the offspring $\mathbf{Y}'$ is updated with $\mathbf{Y}' = \mathbf{Y} + M \cdot \mathbf{d}$.

Given a large positive number (M) (e.g. M = 10,000),
  produce a random direction (d);
  set an iteration counter i = 1;
  while ($\mathbf{Y} + M \cdot \mathbf{d}$ **not** feasible) and (i < *Max_iter*) do
    M = random(M);
    $i = i + 1$;
  endwhile
$\mathbf{Y}' = \mathbf{Y} + M \cdot \mathbf{d}$;

Case study

Data and parameter settings

The proposed model and solution algorithm are applied to a real-world logistics network of Changsha City, Hunan Province, China. The sample regional logistics network comprises 129 links, 70 nodes, and 32 OD pairs for freight, as shown in Fig. 8. The basic data of the logistics network are shown in Table 1.

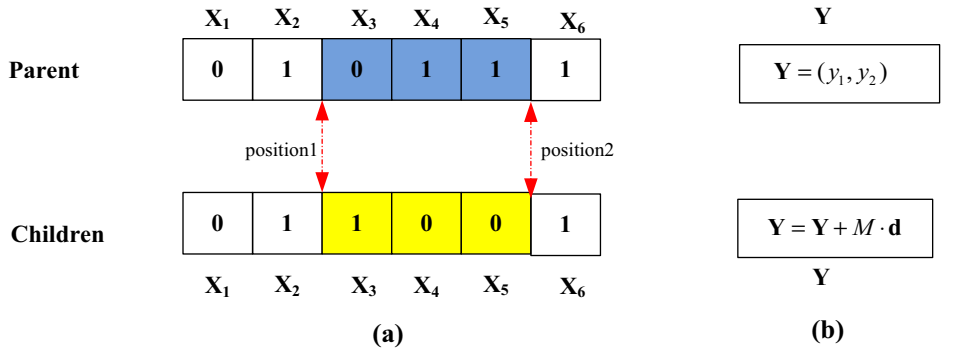


Fig. 7. Representation of the mutation operation.

This logistics network involves two types of logistics demand, namely, industry logistics demand and commercial logistics demand. Given the scarce data on actual logistics OD pairs, we perform rough calculations using an aggregation method to obtain 32 logistics OD pairs, as shown in Table 2. This calculation is based on the data related to the major indicators of the National Economy and Social Development published by the Changsha Statistical Bureau.

Exactly 36 alternative actions are proposed to obtain the carbon reduction targets by using the following strategic decisions: (1) subsidy for green transport modes, such as railways and waterways; and (2) infrastructure improvement, including expanding the capacities of railways, waterways, and road links and logistics nodes. The 36 candidate infrastructure investment projects are shown in Table 3.

The VOT is \$10 per hour (de Jong et al., 2014; Guan and Kazuo, 2000). The average emissions of different transportation modes vary. In particular, the average emissions of HGVs, LGVs, railways, and waterways are 0.132, 0.283, 0.022, and 0.016 (kg/ton km), respectively (Qu et al., 2014). These input data are considered as the base case in the following analysis unless otherwise specified.

Analysis and discussion of results

The proposed solution algorithm was coded in Microsoft Visual C++ 6 and runs on a Dell N5040 laptop with an Intel Pentium 2.13 GHz CPU and 4.00 GB RAM. The number of generations (i.e., stopping criterion) is set to 150. After many simulation runs, we found that the best GA parameter crossover rate (P_c) and mutation rate (P_m) are 0.2 and 0.5, respectively. The iterative process of the GA for the reference case consumes approximately 1296.43 s of CPU time.

Effects of expected minimum carbon-reduction rate targets on model solution

We first compare the solutions of the models using four tests for the different expected minimum carbon reduction rate (EMCRR) targets (10%, 15%, 20%, and 25%), as shown in Table 4 and Fig. 9. Fig. 9 shows that the corresponding BCR value and total cost increase with the increase in the value of the EMCRR targets. The changing processes can be divided into two phases. In the first phase, when the EMCRR rises from 10% to 20%, the total cost slowly increases from \$16,386.58 to \$22,637.32 per week; in the second phase, corresponding to an EMCRR rise between 20% and 25%, the total cost increases sharply from \$22,637.32 to \$28,828.28 per week. This result implies that margin investment increases after the minimum carbon reduction target rate reaches a certain threshold.

Investment and subsidy costs are inclined to rise with the increase in the EMCRR, as shown in Table 4. Furthermore, investing in four candidate projects 5, 7, 28, and 30 is necessary. In other words, developing the transfer capacity of logistics park 49 and logistics park 46, as well as the capacity of the waterway link from node 44 to node 55 should be given priority. The capacity of logistics centers 19, 20, and 29 at the eastern part of the city and that of logistics centers 27 and 31 should be gradually expanded with the increase in the EMCRR.

Effects of the different investment or subsidy schemes

We examine the effects of the different CO_2 emission reduction schemes. The following three schemes are employed: scheme 1 (S1), joint optimization of investments and subsidies; scheme 2 (S2), sole subsidy for railways and waterways; and scheme 3 (S3), sole investment for logistics infrastructure. Table 5 shows the effects of the three different schemes in terms of BCR, total cost, reduction of general logistics cost, percentage of highways, actual CO_2 emission reduction rate, and CO_2 emission per unit of turnover. Table 5 also shows that the total cost, percentage of highways, and CO_2 emission per unit of turnover of scheme 1 are lower than those of the other two schemes. However, the BCR, reduction of general logistics cost, and actual CO_2 emission reduction ratio of scheme 1 are all better than those of schemes 2 and 3. In particular, the total cost of scheme 1 is \$19,313.16 per week, which is 4.19% lower than that of scheme 2 (\$20,158.40 per week) and 8.29% lower than that of scheme 3 (\$21,060.00 per week). The percentage of highways of scheme 1 is 51.25%, which is

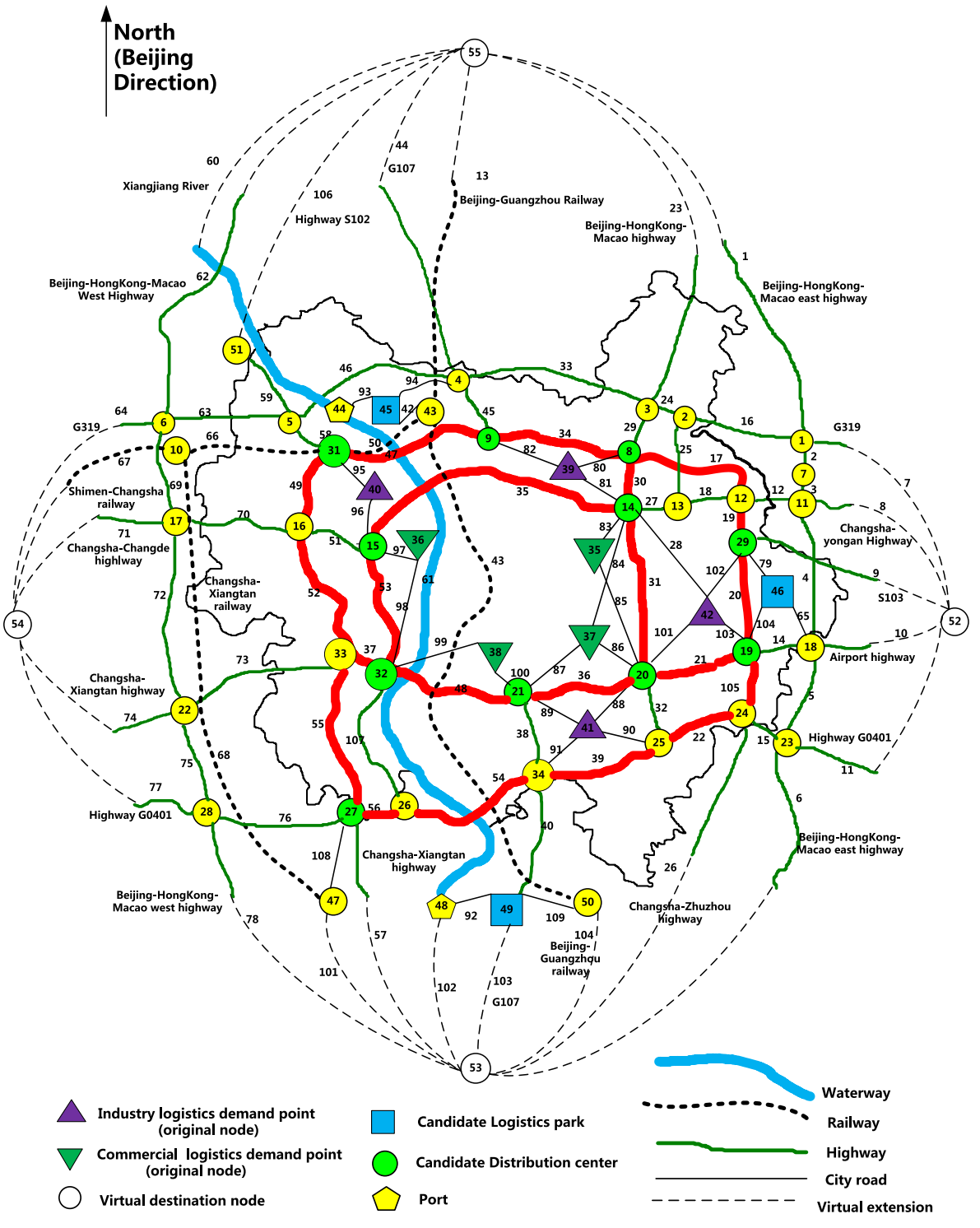


Fig. 8. Logistics network framework of Changsha City, China.

5.13% lower than that of scheme 2 (54.02%) and 4.53% lower than that of scheme 3 (53.68%). The reduction in the general logistics cost of scheme 1 is $10,538.01 per week, which is 5.64% greater than that of scheme 2 ($10,018.72 per week) and nearly the same as that of scheme 3 ($10,572.12 per week). However, the BCR of scheme 1 is 0.548, which is 10.26% greater than that of scheme 2 (0.497) and 9.16% greater than that of scheme 3 (0.502). This finding implies that adopting the joint optimization of investments and subsidies is rational.

Table 1
Basic input data for the logistics network of Changsha City, China. *Source:* Data adapted from references (Department of Commerce of Changsha, 2013; Zhang, 2014).

Arc	From	To	Mode	Length (km)	Time (h)	Unit cost ($/ton km)	Capacity (10,000 tons/week)
1	55	1	2	250.00	3.57	100	60
2	1	7	2	2.00	0.03	0.80	60
3	7	11	2	2.00	0.03	0.80	60
4	11	18	2	9.30	0.13	3.72	60
5	18	23	2	6.67	0.10	2.67	60
6	23	53	2	260.00	3.71	104.00	60
7	52	1	2	156.00	2.23	62.40	60
8	52	11	2	152.00	2.17	60.80	40
9	52	29	2	150.00	2.14	60.00	40
10	52	18	2	155.00	2.21	62.00	40
11	52	23	2	162.00	2.31	64.80	40
12	11	12	2	4.00	0.06	1.60	40
13	55	43	3	250.00	4.17	75.00	80
14	18	19	2	5.00	0.07	2.00	40
15	23	24	2	2.00	0.03	0.80	40
16	1	2	2	8.00	0.11	3.20	40
17	7	8	2	11.67	0.17	4.67	40
18	12	13	2	4.00	0.06	1.60	40
19	12	29	2	3.00	0.04	1.20	40
20	29	19	2	7.00	0.10	2.80	40
21	19	20	2	7.00	0.10	2.80	40
22	24	25	2	6.00	0.09	2.40	40
23	55	3	2	250.00	3.57	100.00	40
24	2	3	2	3.00	0.04	1.20	60
25	2	13	2	6.00	0.09	2.40	60
26	24	53	2	260.00	3.71	104.00	60
27	13	14	2	3.00	0.04	1.20	40
28	42	66	1	8.00	0.20	3.36	60
29	3	8	2	4.00	0.06	1.60	60
30	8	14	2	4.00	0.06	1.60	30
31	14	20	2	11.00	0.16	4.40	30
32	20	25	2	4.00	0.06	1.60	60
33	3	4	2	12.67	0.18	5.07	60
34	8	9	2	9.00	0.13	3.60	30
35	14	15	2	20.00	0.29	8.00	60
36	20	21	2	8.67	0.12	3.47	40
37	32	33	2	2.00	0.03	0.80	30
38	64	34	2	5.33	0.08	2.13	40
39	25	34	2	9.33	0.13	3.73	40
40	34	48	2	10.00	0.14	4.00	40
41	34	49	2	8.33	0.12	3.33	40
42	45	43	0	1.00	0.50	5.00	200
43	43	50	3	40.00	0.67	12.00	80
44	55	4	2	250.00	3.13	100.00	40
45	4	9	2	4.00	0.06	1.60	40
46	4	5	2	11.67	0.17	4.67	40
47	9	31	2	11.00	0.16	4.40	40
48	21	32	2	10.67	0.15	4.27	40
49	31	16	2	6.00	0.09	2.40	30
50	31	43	3	8.00	0.20	3.36	40
51	15	60	1	5.00	0.13	2.10	40
52	16	33	2	9.33	0.13	3.73	30
53	15	33	1	8.00	0.20	3.36	60
54	34	26	2	7.00	0.10	2.80	30
55	33	27	2	11.00	0.16	4.40	30
56	26	27	2	4.00	0.06	1.60	30
57	27	53	2	260.00	3.71	104.00	30
58	5	31	2	3.00	0.04	1.20	30
59	51	5	2	6.00	0.09	2.40	30
60	55	44	4	280.00	7.00	70.00	80
61	44	48	4	41.00	1.03	10.25	120
62	55	6	2	270.00	3.38	108.00	30
63	5	6	2	8.00	0.11	3.20	30
64	6	54	2	163.00	2.33	65.20	30
65	46	18	0	2.00	0.50	10.00	100
66	31	10	3	11.00	0.18	3.30	40

(*continued on next page*)

Table 1 (*continued*)

Arc	From	To	Mode	Length (km)	Time (h)	Unit cost ($/ton km)	Capacity (10,000 tons/week)
67	10	54	3	160.00	2.67	48.00	40
68	10	47	3	32.00	0.53	9.60	30
69	6	17	2	6.00	0.09	2.40	60
70	16	17	2	8.00	0.11	3.20	30
71	17	54	2	155.00	2.21	62.00	30
72	17	22	2	13.00	0.19	5.20	30
73	33	22	2	11.00	0.16	4.40	30
74	22	54	2	165.00	2.36	66.00	30
75	22	28	2	7.00	0.10	2.80	80
76	27	28	2	9.00	0.13	3.60	40
77	28	54	2	155.00	2.21	62.00	40
78	28	53	2	260.00	3.71	104.00	40
79	46	29	0	3.00	0.50	15.00	100
80	56	39	1	4.00	0.10	1.68	80
81	57	39	1	5.00	0.13	2.10	100
82	39	66	1	6.00	0.15	2.52	80
83	66	35	1	4.00	0.10	1.68	40
84	66	37	1	9.00	0.23	3.78	40
85	65	35	1	9.00	0.23	3.78	40
86	65	37	1	4.00	0.10	1.68	40
87	37	64	1	6.00	0.15	2.52	40
88	65	41	1	5.00	0.13	2.10	40
89	41	64	1	5.00	0.13	2.10	40
90	70	41	1	5.00	0.13	2.10	40
91	41	69	1	4.00	0.10	1.68	60
92	49	48	0	2.00	0.50	10.00	100
93	45	44	0	2.00	0.50	10.00	150
94	45	4	0	2.00	0.50	10.00	150
95	40	59	1	4.00	0.10	1.68	80
96	40	15	1	4.00	0.10	1.68	80
97	36	15	1	2.00	0.05	0.84	60
98	36	61	1	8.00	0.20	3.36	60
99	38	61	1	10.00	0.25	4.20	60
100	64	38	1	2.00	0.05	0.84	60
101	42	65	1	5.00	0.13	2.10	80
102	42	67	1	5.00	0.13	2.10	60
103	42	68	1	3.00	0.08	1.26	60
104	46	19	0	1.00	0.50	5.00	40
105	19	24	2	7.00	0.10	2.80	30
106	51	55	2	266.00	3.80	106.40	30
107	32	26	2	9.00	0.13	3.60	30
108	62	47	1	5.00	0.13	2.10	40
109	49	50	0	3.00	0.50	15.00	80
110	47	53	3	260.00	4.33	78.00	30
111	48	53	4	250.00	6.25	62.50	120
112	49	53	2	250.00	3.57	100.00	30
113	50	53	3	250.00	4.17	75.00	80
114	63	48	1	4.00	0.50	20.00	60
115	8	56	0	0.50	0.50	2.50	80
116	9	57	0	0.50	0.50	2.50	80
117	31	58	0	0.50	0.50	2.50	60
118	31	59	0	0.50	0.50	2.50	60
119	16	60	0	0.50	0.50	2.50	60
120	32	61	0	0.50	0.50	2.50	40
121	27	62	0	0.50	0.50	2.50	60
122	27	63	0	0.50	0.50	2.50	150
123	21	64	0	0.50	0.50	2.50	200
124	20	65	0	0.50	0.50	2.50	100
125	14	66	0	0.50	0.50	2.50	60
126	29	67	0	0.50	0.50	2.50	60
127	19	68	0	0.50	0.50	2.50	60
128	34	69	0	0.50	0.50	2.50	100
129	25	70	0	0.50	0.50	2.50	100

Note: In the column labeled as "Mode", "1", "2", "3",and "4", respectively, represent the HGVs, LGVs, Railways, and Waterways, while "0" represents transfer mode by LGVs.

Table 2
Logistics demand OD pairs (10,000 tons/week). *Source:* Data adapted from references (Department of Commerce of Changsha, 2013; Zhang, 2014).

		Origin							
		Commercial logistics demand				Industry logistics demand			
		35	36	37	38	39	40	41	42
Destination	52	19	13	18	11	42	41	28	24
	53	29	19	28	16	64	61	42	35
	54	14	9	14	8	32	31	21	18
	55	34	22	32	19	74	71	49	41

Table 3
Candidate logistics infrastructure investment schemes.

No.	Type	From	To	Capacity (tons/week)	Investment fixed cost ($/week)	No.	Type	From	To	Capacity (tons/week)	Investment fixed cost ($/week)
1	1	45	4	80	1200	19	1	29	46	30	540
2	1	45	44	120	1800	20	2	29	67	30	540
3	1	45	43	50	750	21	2	31	58	40	720
4	1	49	48	60	900	22	2	31	59	40	720
5	1	49	50	60	1080	23	2	32	61	30	540
6	1	46	29	40	720	24	2	34	69	30	540
7	1	46	19	40	720	25	3	10	54	100	3000
8	1	46	18	40	720	26	3	10	47	100	3000
9	2	8	56	30	540	27	3	43	50	100	3000
10	2	9	57	30	540	28	3	47	53	150	4500
11	2	14	66	30	540	29	3	50	53	100	3000
12	2	16	60	30	540	30	3	55	43	150	4500
13	2	19	68	30	540	31	3	58	10	100	3000
14	2	20	65	30	540	32	4	44	48	150	1800
15	2	21	64	30	540	33	4	48	53	200	2400
16	2	25	70	30	540	34	4	55	44	200	2400
17	2	27	62	30	540	35	5	40	15	80	1200
18	2	27	63	30	540	36	5	64	34	60	1200

Note: (1) The estimation of the fixed cost of the logistics infrastructure investment is based on engineering economy theory, which considers the time value of money and recovery period of a project investment.
(2) The fixed cost of the investment is the equivalent uniform per week cost, which is estimated according to the references (National Logistics Standardization Technical Committee of China, 2008; Sullivan et al., 2011; Yong, 2011; Zhang et al., 2015a).

Table 4
Optimal solution of the different expected minimum carbon reduction rate targets.

	10%	15%	20%	25%
Benefit–cost ratio	0.485	0.548	0.590	0.638
Total cost ($/week)	12,386.58	19,313.16	22,637.00	28,828.28
Percent of subsidy (%)	6.95	7.87	10.64	11.22
CO_2 emission per unit turnover (kg/ton km)	0.19	0.17	0.16	0.15
Railway subsidy (%)	2.10	3.06	3.84	7.27
Waterway subsidy (%)	3.20	2.99	6.60	13.24
Investment projects lists	5, 7, 8, 28	3, 5, 6, 8, 12	5, 7, 8, 13, 14	5, 7, 10, 13
	30	13, 14, 17, 18	15, 18, 21, 23	12, 13, 14, 17
		28, 30, 34	28, 30, 34	18, 28, 30, 34

Effect of logistics demand changes on optimal solution

We analyze the effects of the logistics demand on the optimal solution under the EMCRR of 15%. We assume that all logistics demand OD pairs follow the same increasing or decreasing trend under the following change scenarios: 15%, 10%, and 5% decrease; and 5%, 10%, and 15% increase.

Fig. 10 shows that the objective value and total cost increase with the increase in the logistics demand. As shown in Table 6, the logistics infrastructure investment remains at the same level, and the railway or waterway subsidies show subtle fluctuations when the change in all the logistics demand OD pairs falls in the interval [−5%, 5%]. The number of logistics infrastructure projects to invest in decreases from 12 to 11 and 10 when the logistics demand decreases by 10% and 15%, respectively. The corresponding selected projects also change. On the contrary, the number of selected logistics projects rises from 12 to 13 and 14 when the logistics demand increases by 10% and 15%, respectively. This finding indicates the following. (1) The objective value (i.e., BCR value of the logistics system) increases with the gradual increase in logistics demand. (2) Adjusting subsidies to achieve expected minimal CO_2 reduction objectives is feasible when the total logistics demand

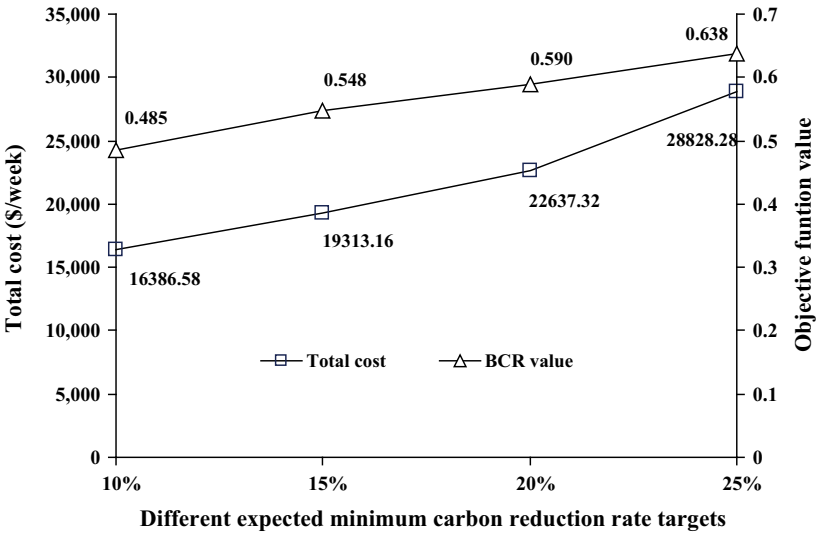


Fig. 9. Effects of expected minimum carbon-reduction rate targets.

Table 5
Effects of different schemes with a given minimum expected carbon reduction rate target (EMCRR = 15%).

	Joint optimization (S1)	Subsidy only (S2)	Investment only (S3)	Change1 (%)	Change2 (%)
Benefit–cost ratio	0.548	0.497	0.502	10.26	9.16
Total cost ($/week)	19313.16	20158.40	21060.00	−4.19	−8.29
Reduction of general logistics cost ($/week)	10538.01	10018.72	10572.12	5.64	0.11
Percent of Highways (%)	51.25	54.02	53.68	−5.13	−4.53
Actual CO_2 emission reduction rate (%)	18.17	16.28	17.51	11.61	3.77
CO_2 emission per unit of turnover	0.169	0.172	0.171	−1.74	−1.17
Railway subsidy (%)	3.06	25.71	–		
Waterway subsidy (%)	2.99	60.82	–		
Investment projects lists	3, 5, 6, 8, 12 13, 14, 17, 18 28, 30, 34	–	4, 10, 15, 16, 18 20, 22, 24, 27 27, 28, 30, 36		

Note: (1) The column labeled as "change1" indicates the percentage of change under the joint optimization scheme (S1) in comparison with that under the subsidy-only scheme (S2).
(2) The column labeled as "change2" indicates the percentage of change under the joint optimization scheme (S1) in comparison with that under the investment-only scheme (S3).

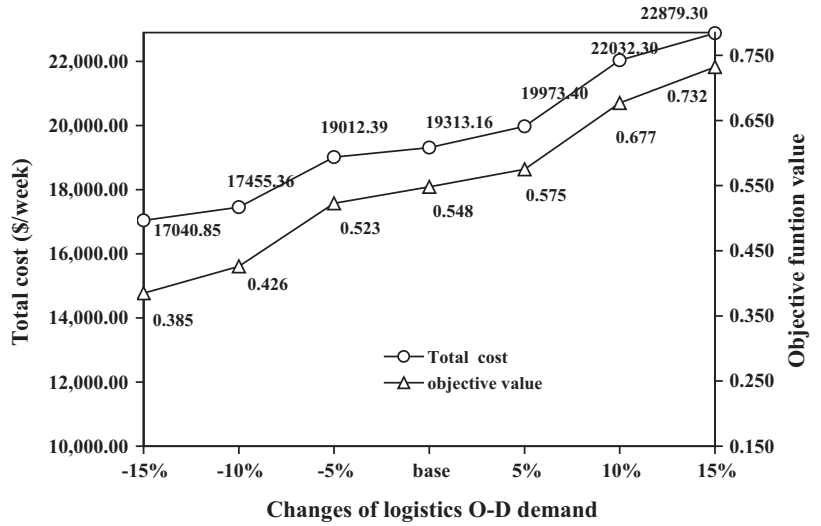


Fig. 10. Effects of changes in logistics demand on optimal solution.

Table 6
Optimal solution of different logistics demands.

	−15%	−10%	−5%	Base	5%	10%	15%
Objective value	0.385	0.426	0.523	0.548	0.575	0.677	0.732
Railway Subsidy	0.100	0.064	0.091	0.031	0.045	0.086	0.042
Water Subsidy	0.091	0.068	0.075	0.030	0.039	0.117	0.101
Total cost	17,040.85	17,455.36	19,012.39	19,313.16	19,973.40	22,032.30	22,879.30
Projects number	10	11	12	12	12	13	14
Invest project list	1, 2, 7, 8 10, 20, 21, 23 27, 30	2, 3, 5, 9 11, 12, 22, 23 30, 32, 36	3, 5, 6, 8 12, 13, 14, 17 18, 28, 30, 34	3, 5, 6, 8 12, 13, 14, 17 18, 28, 30, 34	3, 5, 6, 8 12, 13, 14, 17 18, 28, 30, 34	2, 3, 10, 15 17, 18, 19, 20 21, 28, 30, 32 34	1, 2, 3, 10, 15 17, 18, 19, 20 21, 28, 30, 32 33, 34

maintains fluctuations within the 5% range. (3) Expanding the capacity of green transport links and the logistics transfer of logistics parks is necessary when the increase in logistics demand exceeds 10%.

Conclusion and future studies

In this study, we proposed a new model for the simultaneous optimization of logistics infrastructure investments and state subsidies for green transport modes (such as railways and waterways) to meet the expected carbon emission reduction targets in freight transport. The proposed model was described as a bi-level model. The upper level of the model for the logistics authority was formulated as a mixed integer programming problem aimed at determining the optimal combination of initial schemes for railway and waterway subsidies and improvement actions for a logistics network from among a feasible set of actions. These actions included expanding the capacity of existing logistics nodes (such as logistics parks and distribution centers) and logistics links (i.e., railways, waterways, and highways and city roads). The upper-level model was solved with a GA-based solution algorithm, whereas the lower-level level was formulated as a user equilibrium problem for the selection of logistics service routes.

The model was empirically tested using a real-life case study on the joint optimization of the investments and subsidies in the regional logistics network of Changsha City, Hunan Province, China. The following new insights and important findings were obtained. First, adopting the joint optimization measure that combines logistics infrastructure investments and subsidies for green transport modes is likely to be more effective than adopting them separately under expected carbon reduction targets and budget. Second, different expected carbon reduction target levels can significantly affect the number and size of logistics infrastructure investments and subsidy levels. In particular, high expected carbon reduction target levels equate to a high optimal number and size of logistics infrastructure, as well as to a high total cost ratio, and vice versa.

Opportunities for further research were also identified. First, although the numerical results presented in this paper can be explained logically, case studies on large and realistic logistics networks are necessary to further justify the findings of this work and the performance of the proposed model. Second, the model can be extended as a robust model to address logistics network design with uncertainty in the demand sides, as well as multi-period dynamic investment and pricing problems.

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